

BUSINESS REQUIREMENT DOCUMENT

TicketBoard — Project & Change Request Management Application

Reference Clone Baseline: Zoho Projects (aurionpro instance — "SBI CR" project)

Document Type: Business Requirement Document (BRD)

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Document Control

Revision History

Version	Date	Author	Description of Change
0.1	—	Business Analysis Team	Initial draft derived from reference application walkthrough
1.0	14-Sep-2026	Business Analysis Team	Baseline BRD issued for stakeholder review

Distribution List

Name / Role	Function	Purpose
Product Owner	Business Sponsor	Approval of scope and priorities
Project Manager	Delivery Management	Planning, tracking, resourcing
Business Analyst	Requirements Owner	Requirement elicitation and sign-off
Engineering Lead / Architect	Technical Design	Feasibility, architecture, estimation
QA Lead	Quality Assurance	Test strategy and acceptance validation
UX/UI Designer	Design	Wireframes and visual design

1. Introduction & Purpose

This Business Requirement Document (BRD) defines the functional and non-functional requirements for building “TicketBoard,” a web-based project, task, and issue management application. The requirements in this document are derived from a detailed walkthrough of a live reference implementation (a Zoho Projects instance, tenant-branded “aurionpro,” hosting a real-world project named “SBI CR” — a bank change-request tracking initiative).

The objective is to specify, screen-by-screen and field-by-field, the layout, navigation, data structures, and functional workflows observed in the reference application so that an engineering team can design and build an equivalent, industry-grade product. Every module described here — Projects, Dashboard, Tasks, Issues, Milestones, Time Logs, Reports, and Collaboration — is treated as a first-class requirement area with its own functional specification, data model, and acceptance criteria.

This document is intended to be the single source of truth for Product, Design, Engineering, and QA teams during discovery, design, build, and testing phases of the TicketBoard program.

2. Scope

2.1 In Scope

- Global application shell: left-hand collapsible navigation, top utility bar, notifications, quick-add, search, and account menu.
- Project workspace management: creation, listing, filtering, and summary metrics (Active Projects / Public Projects).
- Project-level modules: Dashboard, Tasks, Reports, Documents, Milestones, Time Logs, and Issues, each accessible via a persistent project sub-navigation bar.
- Task management: hierarchical tasks and subtasks, task detail record with rich metadata, comments, attachments, dependencies, status timeline, and activity stream.
- Issue management: issue records linked to tasks, severity/priority/classification taxonomy, resolution workflow, and comment threads.
- Milestone tracking and association with tasks/issues.
- Time logging: planned vs. logged work hours, billing type classification, and variance reporting.
- Reporting & analytics: a configurable custom-report builder across Tasks, Issues, Time Logs, and Milestones, with chart and tabular views.
- Project & portfolio dashboard: real-time widgets (status charts, KPI tiles, overdue counters).
- Third-party integration: outbound task synchronization to Jira (or equivalent issue tracker).
- Role-based access control and audit-traceable activity history.

2.2 Out of Scope (Phase 1)

- Native mobile applications (a responsive web experience is in scope; dedicated iOS/Android apps are not).
- Billing/invoicing and payment gateway functionality beyond capturing a “Billing Type” attribute.
- Resource capacity planning / Gantt-based resource levelling (may be considered in a later phase).

- Public-facing client portals outside the “Public Projects” listing concept.
- Real-time chat / video conferencing (Collaboration module is limited to comments, mentions, and activity feeds in Phase 1).

3. Stakeholders & Business Objectives

3.1 Key Stakeholders

Stakeholder	Interest / Responsibility
Delivery / Program Managers	Track project health, milestones, and delivery timelines across multiple engagements
Business Analysts	Author and manage change requests, requirement tasks, and BRD-linked artifacts
Developers / Technical Owners	Execute tasks, log time, update statuses, raise/resolve issues
QA Engineers	Raise, track, and resolve issues; verify task completion; manage UAT/SIT exit criteria
Client Stakeholders (e.g., Bank)	Review change-request progress, approve sign-offs, view read-only reports
Executives / Leadership	Consume portfolio-level dashboards and status reports for decision-making

3.2 Business Objectives

1. Provide a single system of record for all project tasks, issues, milestones, and effort tracking, replacing spreadsheet- and email-based tracking.
2. Improve delivery visibility through real-time dashboards and configurable reports for both internal teams and client stakeholders.
3. Standardize the change-request (CR) lifecycle — from requirement gathering through SD sign-off, development, SIT/UAT, and Go-Live — using consistent status taxonomies.
4. Reduce duplicate effort by linking issues directly to the tasks and releases they affect.
5. Enable downstream synchronization with engineering issue trackers (Jira) without duplicate manual entry.
6. Establish auditability: every status change, comment, and field update is traceable via an activity stream.

4. Application Overview & Information Architecture

TicketBoard follows a two-tier navigation model: a Global Navigation Rail (persistent across the whole application) and a Project Sub-Navigation Bar (contextual to the currently open project workspace).

4.1 Global Navigation Rail

The left-hand rail is the primary wayfinding element of the application. It is organized into three zones:

- Primary Navigation: Home, Reports, Projects (default landing module), Collaboration.
- Overview (project-scoped quick links, shown when a project is open): Tasks, Issues, Milestones, Time Logs — each with a search and quick-add icon in its section header.
- Recent Projects: a pinned, recency-ordered list of workspaces the user has visited, each represented by a project icon and name, with sort/filter and search controls.

The top utility bar (present on every screen) provides: tenant branding/workspace switcher, global search, notifications bell with unread badge count, recent-activity clock icon, settings gear, and the user's account avatar/menu.

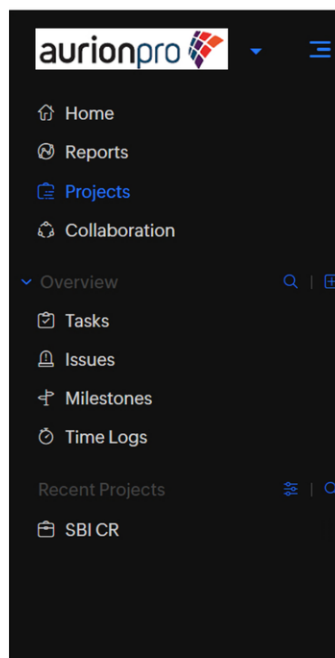


Figure 4.1 — Global navigation rail: primary navigation, Overview quick-links, and Recent Projects.

4.2 Project Sub-Navigation Bar

Once a project workspace is opened (e.g., project “SBI CR”, ID “AP-586”), a horizontal tab bar replaces the generic header and persists across every module screen within that project:

- Dashboard — project-level KPI and chart widgets (landing tab).
- Tasks — hierarchical task list and detail records.
- Reports — custom report builder scoped to the project.
- Documents — file repository associated with the project.

- Milestones — milestone list and status tracking.
- Time Logs — logged effort against tasks.
- Issues — defect/issue register (reached via an overflow “...” menu when tab space is constrained).

The project header additionally displays the project ID/code (e.g., AP-586), project name, a context menu (“...”) for project-level actions, and a refresh/reload control. This pattern must be reused identically on every module screen so users always know which project workspace they are in.

4.3 Consistent List-View Pattern

Every entity list (Projects, Tasks, Issues, Milestones, Time Logs) shares a common interaction pattern that must be implemented as a reusable component:

- A view-switcher (e.g., “List”) with alternate groupings (e.g., “Group By: Task List”, “General” bucket).
- Column sort indicators, per-column filter icons, and a global filter (funnel icon) plus overflow actions menu (“...”).
- A configurable/reorderable column set with a column-chooser affordance at the right edge of the header row.
- Inline progress bars/percentages rendered directly inside table cells (e.g., completion percentage, tasks-closed ratio).
- A primary call-to-action button in the top-right of the list (e.g., “Add Task”) styled as a solid accent-color button.
- A footer showing Total Count and pagination controls.
- Row-level quick actions revealed on hover (open, expand/pop-out, duplicate).

5. Functional Requirements by Module

5.1 Module: Projects

Purpose: Serve as the entry point for all project workspaces, giving users an at-a-glance portfolio view before drilling into any single project.

5.1.1 Projects List Screen

Route: /projects. Tabs: “Active Projects” (default) and “Public Projects”. A dropdown (“All Projects”) allows filtering by status/owner/category.

Required columns (each independently sortable and toggleable):

Column	Description
ID	Auto-generated project code (e.g., AP-586), links to the project workspace
Project Name	Free-text project title; clickable to open the workspace
% (Completion)	Overall completion percentage rolled up from task completion
Owner	Avatar + name of the project owner
Status	Colored pill: Active, On Hold, Completed, Archived, etc.
Tasks	Stacked bar showing closed vs. total tasks (e.g., “50 / 198”) with a hover tooltip (“50 Closed Tasks”)
Milestones	Count or “No Milestones” placeholder
Issues	Stacked bar showing resolved vs. total issues (e.g., “162 / 271”)
Start Date / End Date	Planned project timeline
Tags	Free-form categorization labels

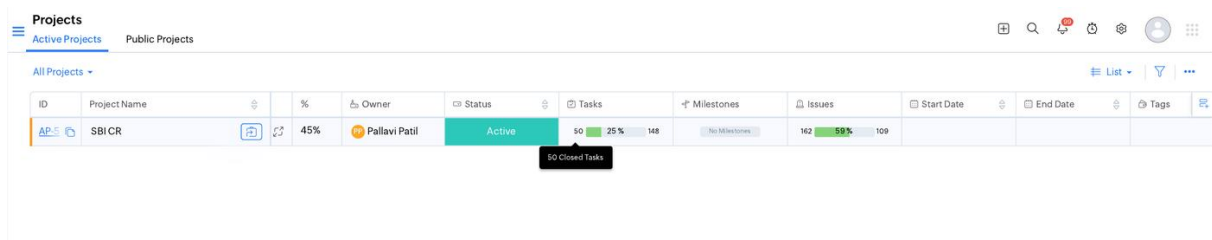


Figure 5.1 — Projects list view showing rollup metrics for tasks and issues with a hover tooltip.

Functional rules:

- Clicking a project row (or its ID) opens the project workspace at the Dashboard tab.
- Tasks/Issues progress bars are color-coded (e.g., green fill for closed proportion) and show an exact tooltip count on hover.
- The list supports horizontal scroll when the column set exceeds viewport width, with a persistent scrollbar at the bottom.
- A footer shows “Total Count: N” with first/prev/next/last pagination controls.

- An “+” icon in the top-right opens a “Create Project” form (name, owner, start/end date, template, tags).

5.2 Module: Dashboard

Purpose: Give project stakeholders an immediate, visual summary of task and issue health without navigating into list screens. This is the default tab when a project is opened.

5.2.1 Layout

A responsive, card/widget-based canvas with an “Add Widget” action and a per-widget overflow menu, arranged as follows:

- KPI Tiles (top row): Total Issues, Open Issues, Overdue Tasks, Open Tasks Due Today — each a compact stat card with a bold numeric value.
- Task Status Report (pie chart): task counts broken down by status (Closed, Open, BRD, Dev in progress, SD Signoff Awaited, SIT in progress, On Hold by Bank, ...), with a scrollable legend.
- Task Status Report (bar chart): identical status breakdown rendered as a horizontally scrollable bar chart with value labels above each bar.
- Issue Status Report (bar chart): issue counts by status (Closed, Open, To..., Not..., Resolved, Won't Fix, Reopened, Hold, ...), with total issue count displayed as a subtitle.

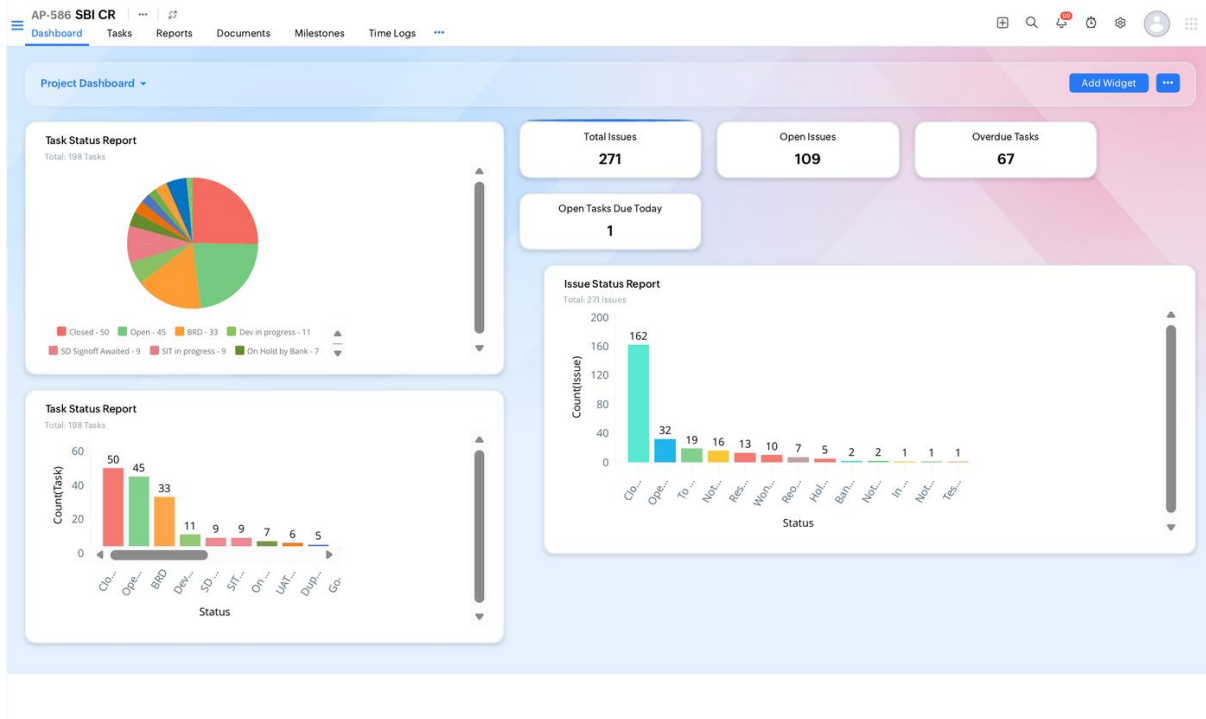


Figure 5.2 — Project dashboard: KPI tiles plus Task Status and Issue Status widgets.

Functional rules:

- Every chart widget must expose its underlying total (e.g., “Total: 198 Tasks”) as a subtitle for context.
- Widgets are independently scrollable/resizable and can be removed or reconfigured via “Add Widget” and the widget overflow menu.
- KPI tiles must be clickable, deep-linking into the corresponding filtered list (e.g., “Overdue Tasks” opens the Tasks list pre-filtered to overdue items).
- Dashboard data must refresh at load and support a manual refresh action.

5.3 Module: Tasks

Purpose: The operational core of the application — tracking discrete units of work (including CR-level parent tasks and their subtasks) through a defined lifecycle.

5.3.1 Task List Screen

Route: /projects/{id}/tasks. Default view groups tasks under a “General” bucket with an expandable count (e.g., “General (148)”). A “Group By” control (default: Task List) allows alternate groupings (by Owner, by Status, by Milestone, by Tag).

ID	Task Name	Associated Team	Status	Task Owner	Start Date	Duration	Priority	Created By	% Completion Percentage
SC6-T443	CR#1: Display of Total amount of Selected Files - Requirement to display...	Not Assigned	Go Live to be p...	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T444	CR#1: Auto Collections BILLING Failure alert to client and CMP	Not Assigned	Blocked	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T445	CR#1: Auto Collections BILLING Failure alert to client and CMP	Not Assigned	Duplicate	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T446	CR#1: Functionality for understanding of NE	Not Assigned	On Hold by Bank	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T447	CR#1: For accounting UPI VM collection transactions at EOD (Meeting)	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T448	CR#1: UPI VM code generation	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T449	CR#1: Option for both transaction password and / or OTP - corporate sa...	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T450	CR#1: Chemical postcard - K2M-A1	Not Assigned	Go Live to be p...	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T451	CR#1: Configuration of transaction count per updated file	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T452	CR#1: Corporate Query Alerts	Not Assigned	On Hold by Bank	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T453	CR#1: Corporate pending charges alert	Not Assigned	Clarification...	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T454	CR#1: EPOD Alerts	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T455	CR#1: EPOD name Validation	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T456	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T457	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T458	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T459	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T460	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T461	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T462	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T463	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T464	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T465	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T466	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T467	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T468	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%
SC6-T469	CR#1: ECL M3 report	Not Assigned	UAT in progress	sample.ashu...	10/06/2024	29 days	Medium	sample.ashu...	0%

Figure 5.3a — Task list (configuration A): ID, Task Name, Associated Team, Status, Task Owner, Start Date, Duration, Priority, Created By, Completion %.

ID	Task Name	Allocated BA	Due Date	Associated Team	Status	Task Owner
SC6-T443	CR#1: Display of Total amount of Selected Files - Requirement to display...	sample.ashu...	10/06/2024	Not Assigned	Go Live to be p...	sample.ashu...
SC6-T444	CR#1: Auto Collections BILLING Failure alert to client and CMP	sample.ashu...	10/06/2024	Not Assigned	Blocked	sample.ashu...
SC6-T445	CR#1: Auto Collections BILLING Failure alert to client and CMP	sample.ashu...	10/06/2024	Not Assigned	Duplicate	sample.ashu...
SC6-T446	CR#1: Functionality for understanding of NE	sample.ashu...	10/06/2024	Not Assigned	On Hold by Bank	sample.ashu...
SC6-T447	CR#1: For accounting UPI VM collection transactions at EOD (Meeting)	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T448	CR#1: UPI VM code generation	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T449	CR#1: Option for both transaction password and / or OTP - corporate sa...	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T450	CR#1: Chemical postcard - K2M-A1	sample.ashu...	10/06/2024	Not Assigned	Go Live to be p...	sample.ashu...
SC6-T451	CR#1: Configuration of transaction count per updated file	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T452	CR#1: Corporate Query Alerts	sample.ashu...	10/06/2024	Not Assigned	On Hold by Bank	sample.ashu...
SC6-T453	CR#1: Corporate pending charges alert	sample.ashu...	10/06/2024	Not Assigned	Clarification...	sample.ashu...
SC6-T454	CR#1: EPOD Alerts	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T455	CR#1: EPOD name Validation	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T456	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T457	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T458	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T459	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T460	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T461	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T462	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T463	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T464	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T465	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T466	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T467	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T468	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...
SC6-T469	CR#1: ECL M3 report	sample.ashu...	10/06/2024	Not Assigned	UAT in progress	sample.ashu...

Figure 5.3b — Task list (configuration B): ID, Task Name, Allocated BA, Due Date, Associated Team, Status, Task Owner.

Required task-list columns (superset, user-configurable per view):

Column	Description
ID	Task code, e.g., SC6-T443; a leading “+” icon indicates the task has subtasks

Column	Description
Task Name	Task title, typically prefixed with a CR reference (e.g., “CR#3 - ...”)
Allocated BA	Business Analyst responsible for requirement clarification
Associated Team	Team assigned for execution, or “Not Associated”
Task Owner	Individual currently responsible for the task
Status	Workflow state pill (see §5.3.3 Task Status Taxonomy)
Start Date / Due Date	Planned dates; overdue due dates render in red with “(N day(s) ago)”
Duration	Computed span in days between start and due date
Priority	P1, P2.1, P2.2, or None, each with a flag icon
Tags	Batch/wave labels (e.g., Top45, Batch4, Batch5.1) plus an overflow “+N” indicator
Created By	Avatar + name of the task creator
Completion Percentage	Inline progress bar with numeric percentage

List functional rules:

- Rows support multi-select via checkboxes for bulk actions (bulk status update, bulk reassignment).
- A “+” prefix on the ID expands an inline subtask tree without leaving the list.
- Clicking a row title opens the Task Detail panel as a right-hand slide-over/modal (see §5.3.2), preserving the list underneath.
- “Add Task” (primary button, top-right) opens a quick-create form; “All Open” dropdown filters by lifecycle state.

5.3.2 Task Detail Screen

Opens as an overlay panel with tab navigation. Header shows: task title (editable inline), breadcrumb chip (Task / task-ID), creator (“By ...”), project chip, comment count, attachment count, a running time tracker (00:00:00) with a start/stop control, and window controls (pop-out to full page, expand, close).

CR#3 - Display of Total amount of Selected Files – Requirement to display the total amount of selected files at the authorization level

By Puller Patel | SBI CR | 00:00:00 | Go-Live to be planned

Description

File	Task	Completion Date
Requirement Analysis & gathering the relevant information		
Single Payment Request, Bulk Payment Request, Over Account transfer		
Secure amount fields to prevent tampering for 30 responses for each record		
Secure transaction event on checkbox click		
Total Amount display – show "Total Amount" value at bottom left		05-Jan-26
Bottom row (same line as Authorize All / Reject All) hide when 0 records selected		
Unit Testing		

Task Information

Associated Team: Sample Shash.com

Task Owner: Pankaj Patel

Allocated BA: SBI CR

Owner: SBI CR

Work Hours: 33:30

Status: Go-Live to be planned

Start Date: 05/01/2026

Due Date: 26/02/2026

Dev Exit: 09/04/2026

SIT Exit: 21/02/2026

UAT Exit: 26/02/2026

SD Delivery: 21/02/2026

Dev Efforts: 3 Days

QC Efforts: 3

Cursor Efforts: 3

Duration: 39 days

Go-Live: 0

Priority: Medium

Completion Percentage: 0

Tags: CR#3 CR#3 CR#3

Reminder: None

Billing Type: None

Recurrence: None

Jira Integration

Create Task in Jira: ☐

JIRA Task Id:

Status of Creation Jira: Not Created

Task Creation Status:

Comments (0) | Subtasks (0) | Log Hours | Documents (0) | Dependency | Status Timeline | Issues | Activity Stream

SBI CR
Awaiting SD certificate from BANK

Figure 5.4 — Task detail: Description, Task Information, and Jira Integration panels.

Panel sections:

Field Group	Fields
Status Bar	Current lifecycle status shown as a prominent dropdown pill directly under the title (e.g., “Go-Live to be planned”)
Description	Rich-text/tabular requirement description; supports embedded tables (e.g., a step-by-step requirement breakdown with a Completion Date column) and an edit/copy affordance
Task Information	Associated Team, Task Owner, Allocated BA, Owner (project chip), Work Hours (planned), Status, Start Date, Due Date, Dev Exit, SIT Exit, UAT Exit, SD Delivery, Dev Efforts, QC Efforts, Cursor Efforts, Duration, Go-Live date, Priority, Completion Percentage, Tags, Reminder, Recurrence, Billing Type
Jira Integration	“Create Task in Jira” checkbox, JIRA Task Id (read-only, populated post-sync), Status of Creation Jira (Not Created / Created / Failed), Task Creation Status log
Tabbed Sub-panels	Comments (threaded, rich-text, count badge), Subtasks (count badge, tabular sub-list with its own Status/Owner/Duration/Dates and “Add Subtask”), Log Hours, Documents (count badge, file attachments), Dependency (predecessor/successor links), Status Timeline (chronological status-change audit trail), Issues (linked issue register), Activity Stream (full field-level change log)

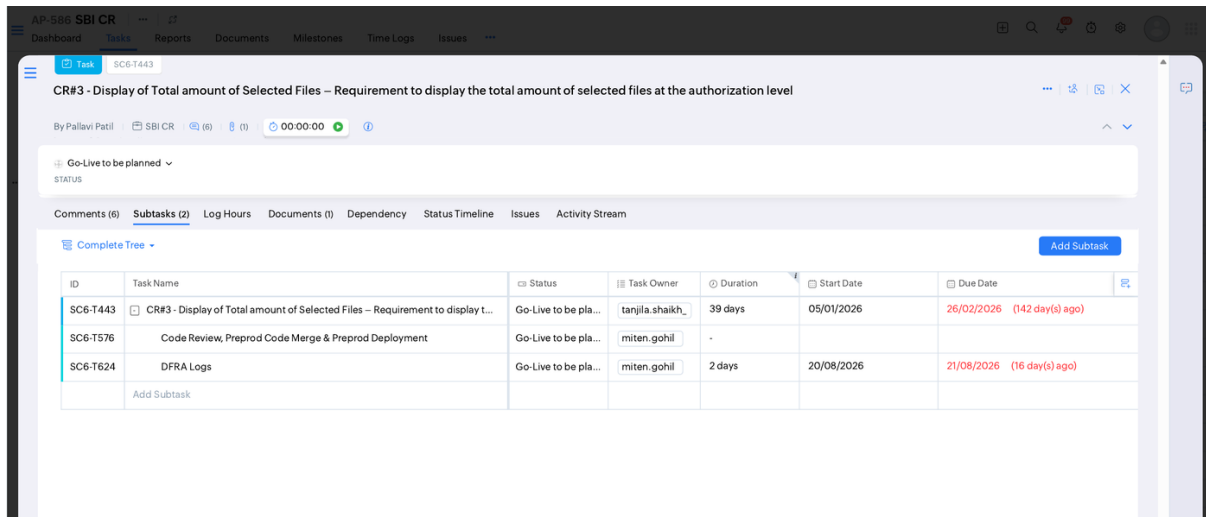


Figure 5.5 — Task detail, Subtasks tab: nested subtask table with its own status, owner, duration, and due-date tracking.

Functional rules:

- The Subtasks tab supports “Complete Tree” expansion and inline “Add Subtask” without leaving the panel.
- Work Hours (planned) is compared against logged time (Time Log Total) with a computed Difference column, surfaced both in the Time Logs report and on the task record.
- All date fields that are past due render in red with a relative “(N day(s) ago)” annotation; upcoming due dates show “(N day(s) to go)”.
- Changing Status updates the Status Timeline automatically and is the trigger event for downstream dashboard/report recalculation.
- Enabling “Create Task in Jira” queues an outbound sync; the panel must reflect an eventual-consistency state machine: Not Created → Pending → Created/Failed.

5.3.3 Task Status Taxonomy (observed)

The reference application enforces a CR-delivery lifecycle. TicketBoard must support a configurable status taxonomy, seeded with the following default workflow states:

Status	Observed Count	Stage
Closed	50	Complete
Open	45	Intake
BRD	33	Requirement definition
Dev in progress	11	Build
SD Signoff Awaited	9	Design sign-off
SIT in progress	9	System Integration Testing
On Hold by Bank	7	Blocked — external dependency
UAT in progress	6	User Acceptance Testing
Duplicate	5	Closed — non-actionable
Go-Live to be planned	4	Release planning

Status	Observed Count	Stage
Blocked	3	Blocked — internal dependency
SD in progress	3	Design
SD to start / SD walkthrough / SIT Exit / UAT Exit / UAT to start / Dev to start / SIT to start / Clarification required from Bank	1–2 each	Various sub-stages

Note: statuses must be configurable per project template (e.g., a simple Kanban template of To Do / In Progress / Done vs. the elaborate CR-delivery workflow shown above) rather than hard-coded, to support reuse across different engagement types.

5.4 Module: Issues

Purpose: Track defects, clarifications, and change-scope questions discovered during any phase, and link each one to the task/feature it affects.

5.4.1 Issues List + Detail (combined workspace)

The Issues screen uses a master-detail layout: a scrollable issue list occupies the left rail (each row showing ID, status pill, truncated title, and reporter), and the selected issue's full record occupies the right pane.

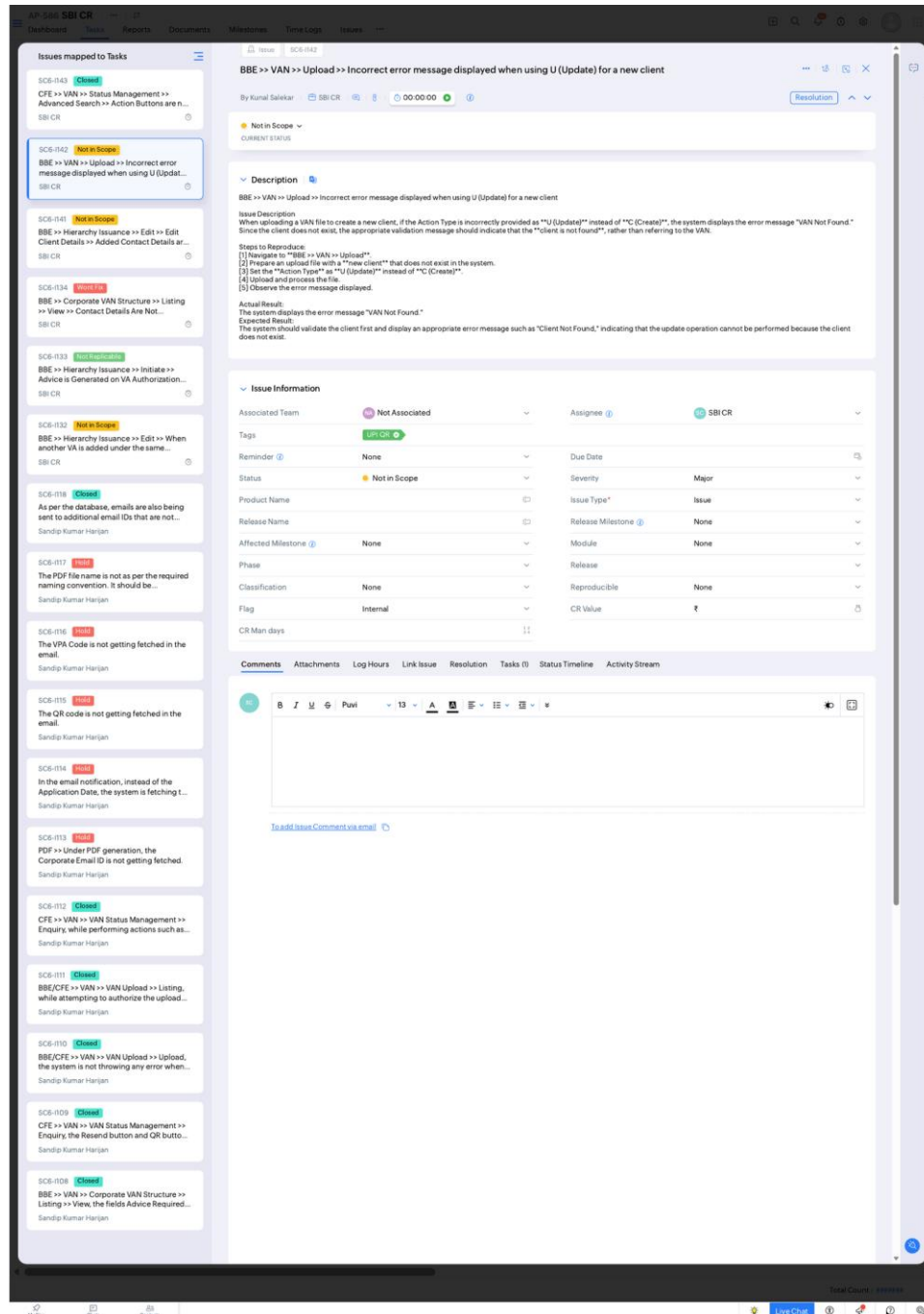


Figure 5.6 — Issues workspace: issue list (left) and issue detail record (right), showing description, reproduction steps, and issue information fields.

Issue detail fields:

Field Group	Fields
Header	Issue title, breadcrumb (Issue / issue-ID), reporter (“By ...”), project chip, elapsed timer, Resolution button, prev/next navigation
Current Status	Prominent status banner (e.g., Not in Scope, Closed, Hold, Won't Fix, Not Replicable)
Description	Structured free text: Issue Description, Steps to Reproduce (numbered), Actual Result, Expected Result
Issue Information	Associated Team, Assignee, Tags, Reminder, Due Date, Status, Severity (Major/Minor/Critical), Product Name, Issue Type, Release Name, Release Milestone, Affected Milestone, Module, Phase, Release, Classification, Reproducible, Flag (Internal/External), CR Value (currency), CR Man-days
Tabbed Sub-panels	Comments (with “add via email” support), Attachments, Log Hours, Link Issue (relate to other issues), Resolution, Tasks (linked task, count badge), Status Timeline, Activity Stream

Functional rules:

- Every issue must be linkable to one or more tasks (the Task tab shows a count badge and the linked task record).
- Issue status changes must be independently timestamped from the parent task's status, with its own Status Timeline.
- Severity and Classification are required fields at creation to support downstream reporting (Issue Basic Reports → Severity, Classification).
- Comments support inbound creation via email (“To add Issue Comment via email”), requiring an inbound-email-to-comment integration.
- The issue list panel must support the same sort/filter/group affordances as the Tasks list.

5.5 Module: Milestones

Purpose: Represent major delivery checkpoints (e.g., release dates, go-live gates) that tasks and issues can be associated with or marked as “Affected.”

- Milestone record fields: Name, Owner, Status, Start Date, Due Date, Description, Flag (Release Milestone vs. Affected Milestone), and roll-up counts of linked tasks/issues.
- Milestone Basic Reports must include Status and Owner breakdowns; Milestone Advanced Reports must include a “Milestone Status by Owner” cross-tab and a “Blueprint Transition Overall Usage Report” for workflow-template adoption tracking.
- Projects lacking milestones must render a “No Milestones” placeholder state in the Projects list (see Figure 5.1).

5.6 Module: Time Logs

Purpose: Capture actual effort against tasks and compare it to planned Work Hours for utilization and billing analysis.

Completion Date	Work Hours (Planned)	Timelog Total	Difference (Planned - Total)	Billing Type
	331.30		+ 331.30	None
	08:00 / day			None
	00:00			None
	161.30		+ 161.30	None
	459.00		+ 459.00	None
	518.30		+ 518.30	None
	17:00		+ 17.00	None
	323.00		+ 323.00	None
	595.00		+ 595.00	None
	433.30		+ 433.30	None
	00:00			None
	195.30		+ 195.30	None
	204.00		+ 204.00	None
	153.00		+ 153.00	None
	986.00		+ 986.00	None
	637.30		+ 637.30	None
	994.30		+ 994.30	None
	773.30		+ 773.30	None
	00:00			None
	85.00		+ 85.00	None
	331.30		+ 331.30	None

Figure 5.7 — Time Log report fragment: planned Work Hours, Timelog Total, computed Difference, and Billing Type per task.

Required fields / columns:

Column	Description
Completion Date	Date the associated task was marked complete (blank while open)
Work Hours (Planned)	Estimated effort, in hh:mm or “N/day” recurring format
Timelog Total	Sum of individually logged time entries against the task

Column	Description
Difference (Planned – Logged)	Computed variance, color-coded (green “+” for hours remaining/under-logged)
Billing Type	Billable / Non-Billable / None — drives invoicing reports

Functional rules:

- Each logged time entry must capture: task reference, user, date, duration, billing type, and an optional note.
- Time Log Basic Reports (Billing Type) and Time Log Advanced Reports (Module; Owner Work Hours by Billing Type) must be available out of the box in the Reports module.
- The system must prevent (or flag) logged time exceeding 24 hours in a single day per user, and support a running per-task timer as seen in the Task/Issue detail header (00:00:00 control).

5.7 Module: Reports

Purpose: A self-service, configurable reporting workbench letting users build, save, and share cross-entity analyses without engineering involvement.

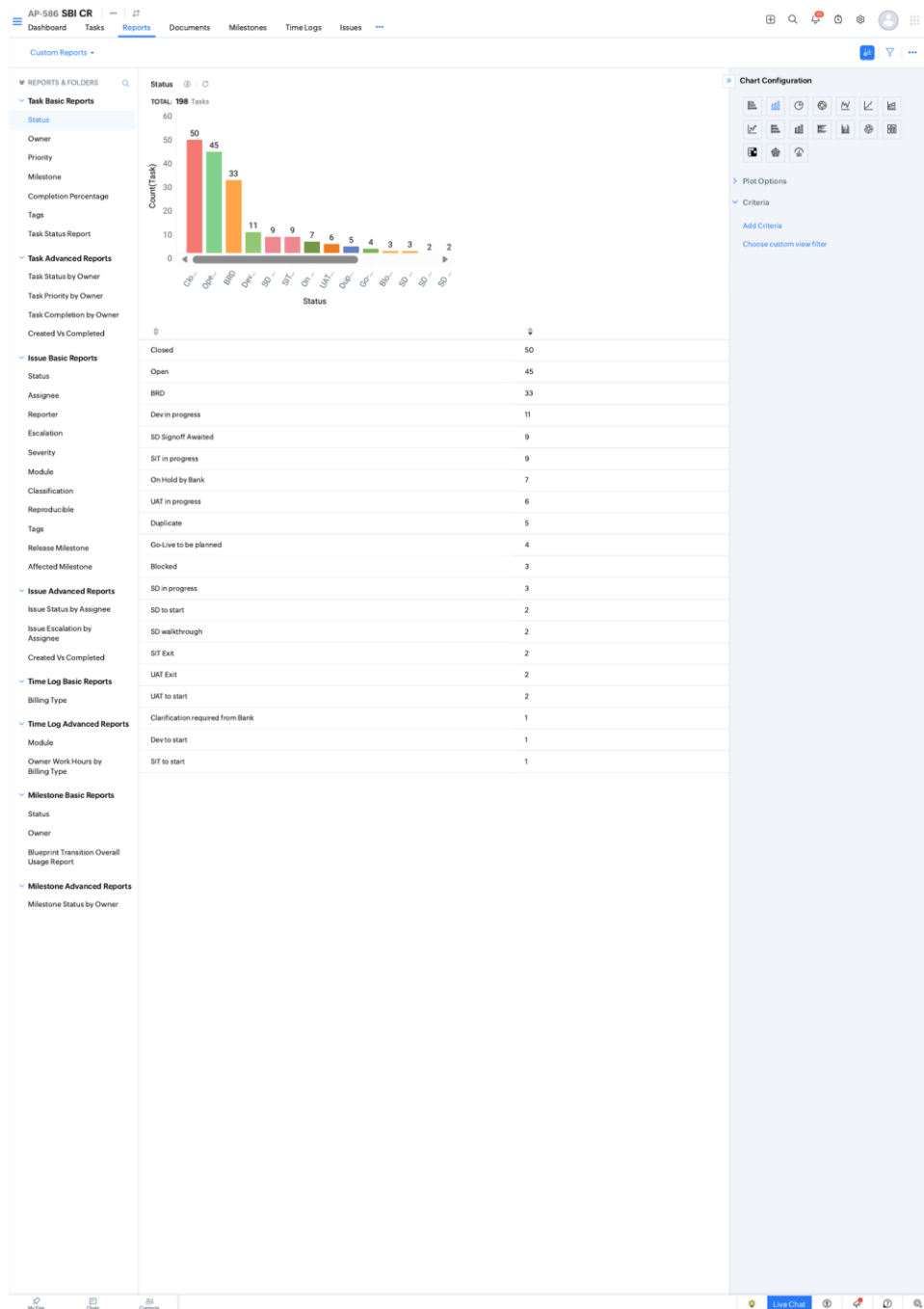


Figure 5.8 — Reports module: report-type navigator (left), chart + tabular result (center), Chart Configuration panel (right).

5.7.1 Report Catalog Structure

Reports are grouped into a left-hand tree, organized by entity and complexity tier:

Category	Included Reports
Task Basic Reports	Status, Owner, Priority, Milestone, Completion Percentage, Tags, Task Status Report
Task Advanced Reports	Task Status by Owner, Task Priority by Owner, Task Completion by Owner, Created vs. Completed
Issue Basic Reports	Status, Assignee, Reporter, Escalation, Severity, Module, Classification, Reproducible, Tags, Release Milestone, Affected Milestone
Issue Advanced Reports	Issue Status by Assignee, Issue Escalation by Assignee, Created vs. Completed
Time Log Basic Reports	Billing Type
Time Log Advanced Reports	Module, Owner Work Hours by Billing Type
Milestone Basic Reports	Status, Owner, Blueprint Transition Overall Usage Report
Milestone Advanced Reports	Milestone Status by Owner

5.7.2 Chart Configuration Panel

A right-hand configuration rail lets the user choose a chart type (bar, line, pie, donut, funnel, scatter, area, radar, and combination variants — presented as an icon grid), set Plot Options (axis, aggregation, sort order), and define Criteria via “Add Criteria” and “Choose custom view filter.”

Functional rules:

- Every report shows a header total (e.g., “TOTAL: 198 Tasks”) and renders both a chart and a paired data table beneath it.
- Reports must support “Custom Reports” creation, saved under a user- or team-defined folder structure (“Reports & Folders”) with search.
- Users must be able to export any report result (CSV/XLSX/PDF) and schedule recurring email delivery (industry-standard expectation, to be confirmed with stakeholders).
- Reports must respect the same field-level and row-level access permissions as the underlying entity records.

5.8 Module: Documents

- Central, project-scoped file repository supporting folder structures, versioning, and attachment linkage from Tasks/Issues (both surface a Documents tab with a live count badge).
- Support drag-and-drop upload, preview for common formats (PDF, images, Office documents), and permission inheritance from the parent project.

5.9 Module: Collaboration

- Team-level activity/feed surface for @mentions, announcements, and cross-project discussion threads, separate from task/issue comments.
- Persistent “My Pins”, “Chats”, and “Contacts” quick-access bar and a “Live Chat” entry point, present at the bottom of every screen for support/collaboration continuity.

6. Functional Flow / Process Sequence

The following end-to-end sequence describes how a unit of work moves through TicketBoard, consistent with the CR-delivery pattern observed in the reference application. This sequence should drive the state-machine and notification design.

7. Project Setup — A Project Owner creates a project workspace (Projects → “+”), defining name, owner, dates, and tags. The project appears in “Recent Projects” and the Active Projects list.
8. Requirement Intake — A Business Analyst logs a Task representing a Change Request (e.g., “CR#3 – ...”), capturing the structured requirement description, Allocated BA, Priority, and initial Status (“BRD”).
9. Design & Sign-off — The task progresses through SD in progress → SD walkthrough → SD Signoff Awaited, with SD Delivery date captured.
10. Development — Status moves to Dev to start → Dev in progress; Dev Efforts and Dev Exit date are recorded; subtasks (e.g., “Code Review, Preprod Code Merge & Preprod Deployment”) are created and tracked independently.
11. Issue Discovery — At any stage, QA or stakeholders raise Issues against the task (e.g., defects found during Dev/SIT/UAT), each carrying Severity, Classification, and Reproducible flags, and linked back to the originating task via the Tasks tab.
12. System & User Testing — Status transitions through SIT to start → SIT in progress → SIT Exit, then UAT to start → UAT in progress → UAT Exit, with corresponding exit dates captured.
13. External Dependency Handling — If blocked by a third party (e.g., “On Hold by Bank”) or internally (“Blocked”), the status reflects this and the Status Timeline logs the transition with a reason/comment.
14. Go-Live Planning & Closure — Status moves to “Go-Live to be planned”, a Go-Live date is set, and upon release the task is marked “Closed” with Completion Percentage = 100%.
15. Effort Reconciliation — Throughout the lifecycle, users log time against the task; Time Logs continuously compute Difference vs. planned Work Hours for utilization reporting.
16. Reporting & Dashboard Refresh — Every status/field change immediately updates the project Dashboard widgets and any saved Reports referencing that entity, and appends an entry to the Activity Stream.
17. Optional Jira Sync — If “Create Task in Jira” is enabled, an outbound integration event creates/updates the corresponding Jira issue and reflects the resulting JIRA Task Id and creation status back on the TicketBoard task record.

7. Data Model — Core Entities

This section defines the minimum entity attributes required to support the functionality specified in Section 5. Primary keys are denoted (PK) and foreign keys (FK).

7.1 Project

Attribute	Type	Notes
project_id (PK)	String	Human-readable code, e.g., AP-586
name	String	Required
owner_id (FK → User)	String	
status	Enum	Active, On Hold, Completed, Archived
completion_pct	Decimal	Derived from task rollup
start_date / end_date	Date	
visibility	Enum	Private / Public (drives “Active Projects” vs. “Public Projects” tab)
tags	Array<String>	

7.2 Task

Attribute	Type	Notes
task_id (PK)	String	e.g., SC6-T443
project_id (FK)	String	
parent_task_id (FK, nullable)	String	Self-referencing for subtasks
title	String	Required
description	Rich Text / Table	Supports embedded tabular requirement steps
status	Enum (configurable)	See Task Status Taxonomy (§5.3.3)
associated_team	String (FK → Team, nullable)	
task_owner_id / allocated_ba_id (FK → User)	String	
priority	Enum	P1, P2.1, P2.2, None
work_hours_planned	Duration	hh:mm or recurring “N/day”
start_date / due_date / dev_exit / sit_exit /	Date	Milestone-linked date fields

Attribute	Type	Notes
uat_exit / sd_delivery / go_live_date		
dev_efforts / qc_efforts / cursor_efforts	Decimal (days)	
completion_pct	Decimal	
tags	Array<String>	Batch/wave labels
billing_type	Enum	Billable / Non-Billable / None
jira_sync_enabled	Boolean	
jira_task_id	String, nullable	Populated post-sync
jira_creation_status	Enum	Not Created / Pending / Created / Failed
created_by_id (FK → User)	String	
created_at / updated_at	Timestamp	

7.3 Issue

Attribute	Type	Notes
issue_id (PK)	String	e.g., SC6-I142
project_id (FK)	String	
title / description	String / Rich Text	Description, Steps to Reproduce, Actual/Expected Result
status	Enum	Open, Closed, Hold, Not in Scope, Won't Fix, Not Replicable, ...
severity	Enum	Critical, Major, Minor
classification / module / phase / release	String	
reproducible	Enum	Yes / No / Intermittent
associated_team / assignee_id (FK)	String	
reporter_id (FK → User)	String	
release_milestone_id / affected_milestone_id (FK → Milestone)	String, nullable	
flag	Enum	Internal / External
cr_value	Currency	Change-request monetary value, if applicable
cr_man_days	Decimal	
linked_task_ids (FK → Task, many)	Array<String>	

Attribute	Type	Notes
due_date / reminder	Date	

7.4 Milestone

Attribute	Type	Notes
milestone_id (PK)	String	
project_id (FK)	String	
name / description	String	
owner_id (FK → User)	String	
status	Enum	Upcoming, In Progress, Completed, Missed
start_date / due_date	Date	
linked_task_ids / linked_issue_ids	Array<String>	Roll-up counts shown on Projects list

7.5 Time Log

Attribute	Type	Notes
timelog_id (PK)	String	
task_id (FK)	String	
user_id (FK → User)	String	
log_date	Date	
duration	Duration (hh:mm)	
billing_type	Enum	Billable / Non-Billable / None
note	String, optional	

8. User Roles & Permissions

The following role matrix is proposed as the default RBAC model, to be validated with stakeholders during design sign-off.

Capability	Admin	Project Manager	Team Member	Client / Viewer
Create / archive projects	Yes	Yes	No	No
Configure workflows & statuses	Yes	Yes (own projects)	No	No
Create / edit / delete tasks	Yes	Yes	Yes (assigned scope)	No
Create / edit issues	Yes	Yes	Yes	Comment only
Log time	Yes	Yes	Yes	No
Build / save custom reports	Yes	Yes	View only	View shared only
Manage users & roles	Yes	No	No	No
Configure Jira integration	Yes	No	No	No
View dashboards	Yes	Yes	Yes	Yes (scoped)

9. Third-Party Integration Requirements

9.1 Jira Integration

Observed in the Task detail record's "Jira Integration" panel. Required behavior:

- A per-task "Create Task in Jira" toggle initiates an outbound sync to a configured Jira project/issue type mapping.
- "Status of Creation Jira" must reflect one of: Not Created, Pending, Created, Failed, and "JIRA Task Id" is populated (read-only, with a copy/deep-link icon) upon success.
- A free-text/log "Task Creation Status" field captures the last sync response or error detail for troubleshooting.
- Sync must be idempotent (re-toggling an already-created task must not create duplicate Jira issues) and retry-capable on transient failures.

9.2 Email-to-Comment Integration

- Both Task and Issue comment threads must support comment creation via inbound email ("To add Issue Comment via email"), matching sender-to-user and thread-to-record via a unique reply-to address or token.

9.3 Notification Channels

- In-app notification bell with unread badge count (observed on the top utility bar) for assignments, mentions, status changes, and approaching due dates.
- Email notification digesting, and (optional, Phase 2) chat-platform (e.g., Slack/Teams) webhook notifications.

10. Non-Functional Requirements

10.1 Performance

- List and dashboard screens must render initial content within 2 seconds for datasets up to 10,000 records per project, with virtualized/paginated rendering beyond that.
- Chart widgets must lazy-load independently so a slow report does not block the rest of the dashboard.

10.2 Usability & Accessibility

- Consistent dark-themed global navigation and light-themed workspace content, matching the visual pattern of the reference application (dark rail, light canvas).
- WCAG 2.1 AA conformance: color is never the sole indicator of status (status pills must also carry text labels); full keyboard navigability of list and detail views.
- Responsive layout: sub-navigation tabs collapse into an overflow “...” menu on narrower viewports, exactly as observed for the project tab bar.

10.3 Security

- Role-based access control enforced at both API and UI layers (see §8).
- All record mutations captured in an immutable Activity Stream / Status Timeline for audit purposes.
- Field-level flags such as “Internal” vs. “External” (Issue → Flag) must be respected by any client-facing/public view.

10.4 Availability & Data Integrity

- Target 99.9% uptime SLA for production; automated daily backups with point-in-time recovery.
- Referential integrity enforced between Tasks ↔ Issues ↔ Milestones ↔ Time Logs (no orphaned links after entity deletion; soft-delete/archive pattern preferred over hard delete).

10.5 Browser & Device Support

- Latest two major versions of Chrome, Edge, Firefox, and Safari on desktop; responsive support down to tablet viewport widths.

11. Acceptance Criteria (Sample — Task Module)

- Given a user with Team Member permissions, when they open a project's Tasks tab, then the list renders grouped under “General” with a live count matching the total task records in that project.
- Given a task with subtasks, when the user clicks the “+” expand icon on its ID, then all subtasks render inline with their own Status/Owner/Duration/Due Date without a full page navigation.
- Given a task whose due date has passed and status is not Closed, when the list renders, then the Due Date cell displays in red with a “(N day(s) ago)” suffix.
- Given a task with “Create Task in Jira” enabled, when the sync completes successfully, then “Status of Creation Jira” updates to “Created” and “JIRA Task Id” is populated within the configured SLA (e.g., 60 seconds).
- Given any change to a task's Status field, when the change is saved, then a corresponding entry appears in the Status Timeline tab and the project Dashboard's Task Status widgets reflect the new counts on next load.
- Given a Time Log entry logged against a task, when the user views the Time Log Advanced Report, then Work Hours (Planned), Timelog Total, and the computed Difference reconcile to the individual entries logged.

12. Assumptions & Constraints

18. The reference application (Zoho Projects, aurionpro tenant) is used strictly as a UX/functional baseline; TicketBoard is a clean-room, independently built product and will not reuse Zoho's proprietary code, branding, or backend.
19. Exact field lists, statuses, and counts documented here (e.g., “198 Tasks”, “271 Issues”) are illustrative of the reference project's live data at the time of capture and are not fixed system limits.
20. Status taxonomies (Task Status Taxonomy, Issue statuses) are assumed to be admin-configurable per project template rather than globally hard-coded.
21. Jira is assumed as the initial third-party integration target; the integration layer should be built against an abstracted interface to allow additional trackers (e.g., Azure DevOps) in later phases.
22. Currency for CR Value fields defaults to INR (₹) based on the reference data; must be made configurable per tenant.

13. Glossary

Term	Definition
BRD	Business Requirement Document — also used in this application as an early task status representing the requirement-definition stage
CR	Change Request — a unit of scoped work requested by a client, typically modeled as a parent Task
SD	Solution Design — the design/specification phase preceding development
SIT	System Integration Testing
UAT	User Acceptance Testing
Blueprint	A reusable workflow/status template applied to milestones or projects
KPI Tile	A compact dashboard widget surfacing a single aggregate metric
Rollup	A computed aggregate value (e.g., completion %) derived from child records